

SQL Project 2

1. Customer Purchase History

Problem Statement: Retrieve all orders made by a specific customer (CustomerID = 11000).

Expected Output:

- Order ID
- Order Date
- Order Amount

Tables: `Sales.SalesOrderHeader`

2. Orders Placed

Problem Statement:

Retrieve all orders that were placed in the year **2011**.

Tables:

- `Sales.SalesOrderHeader`

Columns:

- `SalesOrderHeader.OrderDate`
- `SalesOrderHeader.TotalDue`

3. Most Expensive Products

Problem Statement:

Retrieve the top 15 most expensive products based on the list price.

Tables:

- `Production.Product`

Columns:

- `Product.Name`
- `Product.ListPrice`

4. Specific Region Customers

Problem Statement:

Retrieve customer details who belong to the '**Pacific**' sales territory.

Tables:

- `Sales.Customer`
- `Sales.SalesTerritory`

Columns:

- `Customer.CustomerID`
- `Customer.PersonID`
- `SalesTerritory.Name`

5. Order Count by Customer

Problem Statement: List customers and the number of orders they have placed.

Expected Output:

- Customer ID
- Total Orders

Tables: `Sales.SalesOrderHeader`

6. Distinct Product Categories

Problem Statement:

Retrieve all unique product categories available in the database.

Tables:

- `Production.ProductCategory`

Columns:

- `ProductCategory.Name`

7. Products Never Sold

Problem Statement: Identify products that have never been sold.

Expected Output:

- Product Name
- Product ID

Tables: `Production.Product`,
`Sales.SalesOrderDetail`

8. Unshipped Orders

Problem Statement: Retrieve orders that have not been shipped yet.

Expected Output:

- Order ID
- Order Date
- Ship Date (should be NULL)

Tables: `Sales.SalesOrderHeader`

9. Most Common Order Status

Problem Statement: Identify the most frequent order statuses in the system.

Expected Output:

- Order Status
- Order Count

Tables: `Sales.SalesOrderHeader`

10. Average Order Value by Customer

Problem Statement: Calculate the average order value per customer.

Expected Output:

- Customer ID
- Average Order Value(Total Due)

Tables: `Sales.SalesOrderHeader`

11. High-Value Customers

Problem Statement: Identify customers whose total purchase amount is greater than \$10,000.

Expected Output:

- Customer ID
- Total Purchase Amount

Tables: `Sales.SalesOrderHeader`

12. Top 5 Sales Representatives

Problem Statement: Find the top 5 sales representatives with the highest total sales.

Expected Output:

- Sales Representative ID
- Name
- Total Sales Amount

Tables: `Sales.SalesPerson`, `Person.Person`,
`Sales.SalesOrderHeader`

14. Employee Order Handling Efficiency

Question:

Find the average time taken by each employee to process orders. Include:

- **EmployeeID**
- **Average Processing Time (Days)**

Tables:

- `Sales.SalesOrderHeader`

Columns:

- `SalesPersonID`
- `OrderDate`
- `ShipDate`

15. Delayed Orders Report

Question:

Find orders that took more than 7 days to be shipped. Include:

- **SalesOrderID**
- **Order Date**
- **Ship Date**
- **Days to Ship**

Tables:

- `Sales.SalesOrderHeader`

Columns:

- `SalesOrderID`
- `OrderDate`
- `ShipDate`

16. Product Profitability Analysis

Question:

Find the top 11 most profitable products. Include:

- **ProductID**
- **Product Name**
- **Total Profit**

Tables:

- `Sales.SalesOrderDetail`
- `Production.Product`

Columns:

- `ProductID`
- `UnitPrice`
- `StandardCost`
- `OrderQty`